

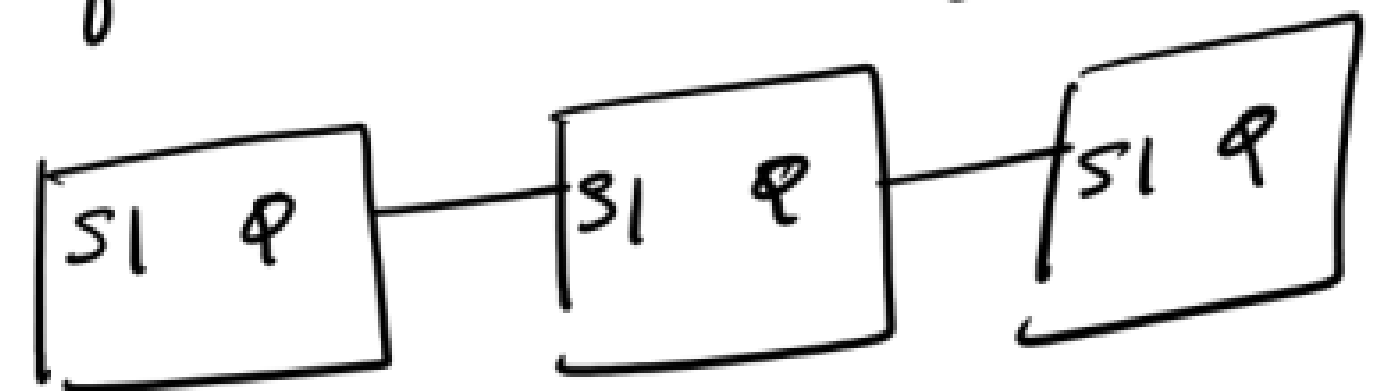
Need of Scan insertion :

Scan Insertion is needed because every node in the design should be toggled to gain controllability and observability of the design and this can help in gaining high test coverage of the entire circuit.

Why we converted DFF $\rightarrow$ SFF?

$\hookrightarrow$ we did it so that we can have an additional scan i/p.

$\hookrightarrow$ if we have scan i/p, we can form shift reg path (q - si path) $\Rightarrow$ scan chain



What is the use of forming the shift reg path?

$\hookrightarrow$ We can easily load the i/p reg into our scan chain.

$\hookrightarrow$ Cap value can be easily loaded out from the scan chain.